

# UK Renewable Energy Dashboard

Design Report (CW-2) - SET09123 Interactive Data Visualisation

**Question:** Which renewable energy sources drove the growth of UK renewable energy from 1990 to 2023, and how did this change the renewable share of total UK energy consumption?

**Dataset:** Office for National Statistics, “Energy use: renewable and waste sources”, dataset 1990 to 2023, released 5 June 2025. Values in Mtoe.

## 1. What I wanted the dashboard to do

I wanted the dashboard to answer one question properly, instead of being a wall of charts. The ONS dataset is interesting because two things are happening at the same time. Renewable use grows a lot, especially after 2005. And total UK energy use actually drops, especially after 2010. Both of these affect the renewable share, and most news coverage I read while researching only showed one side or the other. I wanted to put both sides on the same screen.

The question naturally splits in two. Which sources drove the growth, and what was the outcome on the share. Each half gets a main chart. The other two charts answer the obvious follow-ups, what does the mix look like in a given year, and where in the economy is this energy actually being used.

## 2. From coursework 1 to coursework 2

My coursework 1 prototype had a donut, a multi-line, and a grouped bar. It worked, but it was not really answering anything. I had not picked a question, so the layout had no logic to it. The donut also just looked off because Bioenergy and Wind dominate the mix, and the small slices were impossible to compare side by side.

For coursework 2, I rebuilt around a clear question and made every panel earn its place. Three things changed. First, the donut became a treemap. Rectangles are easier to compare than arcs when slice sizes are very uneven. This also satisfied the brief's requirement for a hierarchical visualisation.

Second, I extended the interactions. In CW1, hovering was the only interaction. Now you can persistently select a source and a year, and they work independently, so you can combine filters. I also added a brush for range selection on the line chart, plus zoom on the line and sector charts.

Third, the layout was rebuilt to fit one screen at 1900 by 900. I added a KPI strip across the top that adapts to whatever you have filtered, and three header badges that show the active filter state. The grouped bar from CW1 became a single-year sector view, and I repositioned it as supporting context rather than a direct answer to the main question.

## 3. Layout and the one-screen rule

The brief asks for no scrolling, and I took that seriously. If you have to scroll, you lose the cross-chart context, which kind of defeats the point of a dashboard. The layout is a 2 by 2 grid below a header strip and a KPI strip. The header has the title, the dashboard question, the year filter dropdowns, the Range select and Reset buttons, and the three filter badges. The KPI strip below sits across the full width and shows four numbers that adapt to the filters.

Top row is the line chart on the left and the treemap on the right. Bottom row is the share chart on the left and the sector chart on the right. I tested at 1366 by 768 and 1920 by 1080. Both fit. I also added a small media query so the rows do not collapse to nothing on shorter windows. Figure 1 shows the default state.

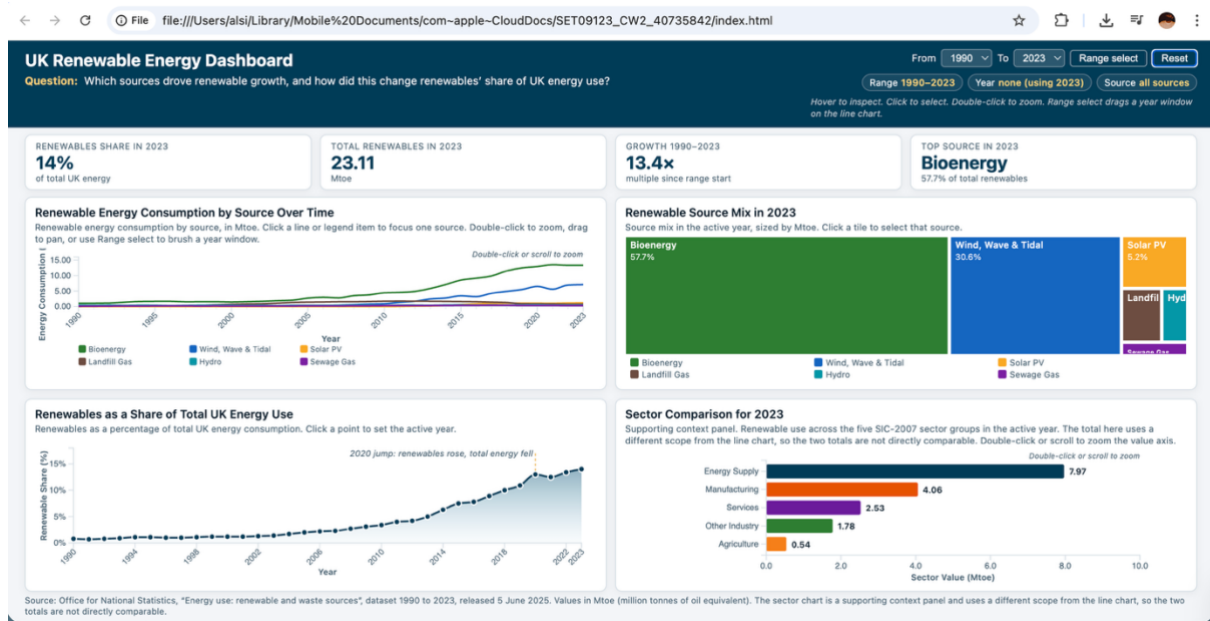


Fig.1- Default view, 1990 to 2023, no selection. KPIs read overall totals.

#### 4. Line chart

The line chart is the main answer to the first half of the question. Six sources, 34 years, all on the same Mtoe scale. I went with a multi-line because it lets you compare all six series at once, and Cleveland and McGill (1984) showed position on a common scale is the most accurate way to read quantitative data. I tried a stacked area chart first because it looks nice, but it actually hides the moment when Wind overtakes everything except Bioenergy in the mid 2010s. So I went back to lines.

Each source has its own colour, and I used the same palette across every chart, so Bioenergy is always green, Wind is always blue, and so on. I checked the palette in a colour-blind simulator, and the sources still read apart at low values. There is also a y-axis rescale when you select a single source. Without it, Solar PV looks completely flat against a Bioenergy-sized scale, because Solar PV peaks at 1.19 Mtoe in 2023 while Bioenergy is at 13.33. With the rescale, the Solar PV growth from near zero is finally readable.

#### 5. Treemap

The treemap shows the source mix for the active year. I picked it over a donut because rectangles read more accurately than arcs when slices are uneven, and Heer, Bostock, and Ogievetsky (2010) make a similar point about treemaps for part-to-whole breakdowns.

Tiles are clickable. Click the Wind tile, and Wind becomes the selected source across every chart. There is also a legend underneath because in the early 1990s Solar PV is so small its tile is basically invisible, and the legend is the only way to identify the colour. For tile labels, I wrote a small fitter that picks the largest font size that fits and falls back to a short name when the tile is too narrow. This took longer than I expected because I first tried estimating text width from character count and font size, but my estimates kept being wrong, so I switched to a hidden SVG measurement element, which is way more reliable. Took me about an hour to figure out the CSS was overriding my font-size attribute, which was annoying.

#### 6. Share chart

The share chart answers the second half of the question. Renewables as a percentage of total UK energy. I combined an area fill, a line, and clickable dots. The area gives a visual sense of size, the line keeps the trend crisp, and the dots are the click target.

There is a small annotation on 2020 because that point matters. The share rose to 13% that year, but a chunk of that rise was actually total UK energy falling during the pandemic, not renewables growing

extra fast. I wanted to flag this because if you only look at the line, you would think renewables had a huge year, when really the pandemic shrank the denominator. Hover shows the year, the renewable share, the year-on-year change in percentage points, and the absolute renewables value behind the percentage.

## 7. Sector chart

In CW1, this panel was a grouped bar with all 34 years and 5 sectors at once. It packed 170 bars into a small area and was honestly not very useful. For CW2, I changed it to a single-year horizontal bar showing where the active year's renewable use sits across the five SIC-2007 sector groups. Bars are sorted descending so the dominant sector reads first. Numbers sit at the end of each bar.

The sector panel is supporting context, not a second view of the same totals. The sector chart uses a different scope from the line chart and treemap, so the totals do not match in the same year. In 2023 the sector total is 16.87 Mtoe while the line chart total is 23.11 Mtoe. I added a note in the dashboard footer about this, so a marker or reader does not assume they should add up. The sector chart is a comparative view of where the use sits, not a balance against the source-side number.

In 2023, Energy Supply leads at 7.97 Mtoe, then Manufacturing at 4.06, Services at 2.53, Other Industry at 1.78, and Agriculture at 0.54.

## 8. Interactions

I followed a simple rule. Hover inspects, click commits. Hover never changes state, only shows a tooltip and a temporary highlight. Click selects, and the selection sticks until you click it again or hit Reset. Pike et al. (2009) talk about this kind of separation as a basic part of interactive visualisation design, and that is roughly the model I followed.

Source selection and year selection are independent. Picking a source does not clear the year, and picking a year does not clear the source. So, for example, if you want to look at Bioenergy in 2020 specifically, you click the Bioenergy tile and then the 2020 dot on the share chart, and both filters stay active together. The header badges turn amber when something is selected, so you always know what is active. Figure 2 shows Bioenergy selected.

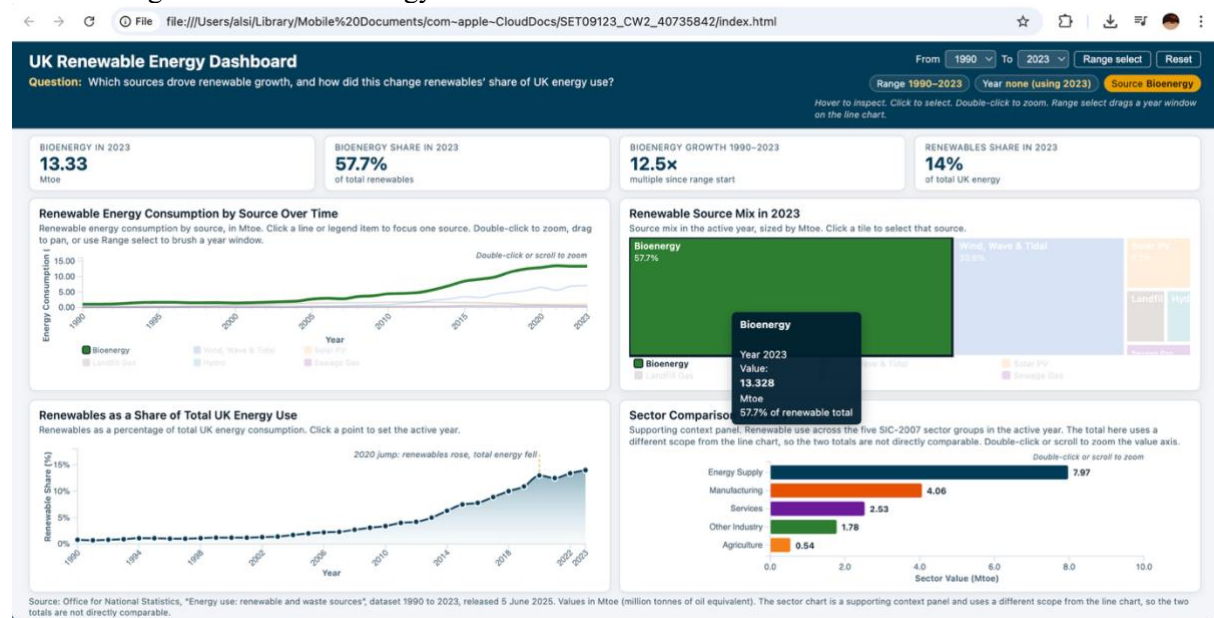


Fig.2- Bioenergy selected. Other sources fade. KPIs switch to Bioenergy specific values.

Fig. 3 shows the year-only state with 2020 selected. Treemap and sector chart rebuild for that year.

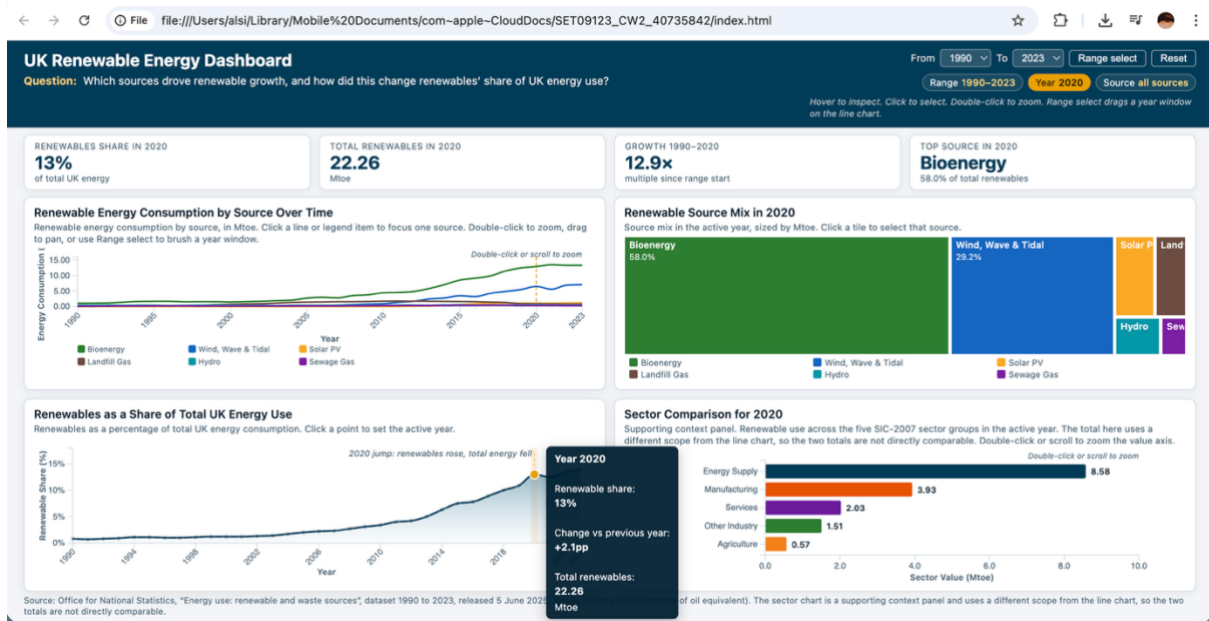


Fig.3- Year 2020 selected. Treemap and sector chart rebuild for 2020.

Fig. 4 is the combined state, Bioenergy in 2020. KPIs answer that specific combination: 12.91 Mtoe, 58% of renewables, 12.1 times growth.

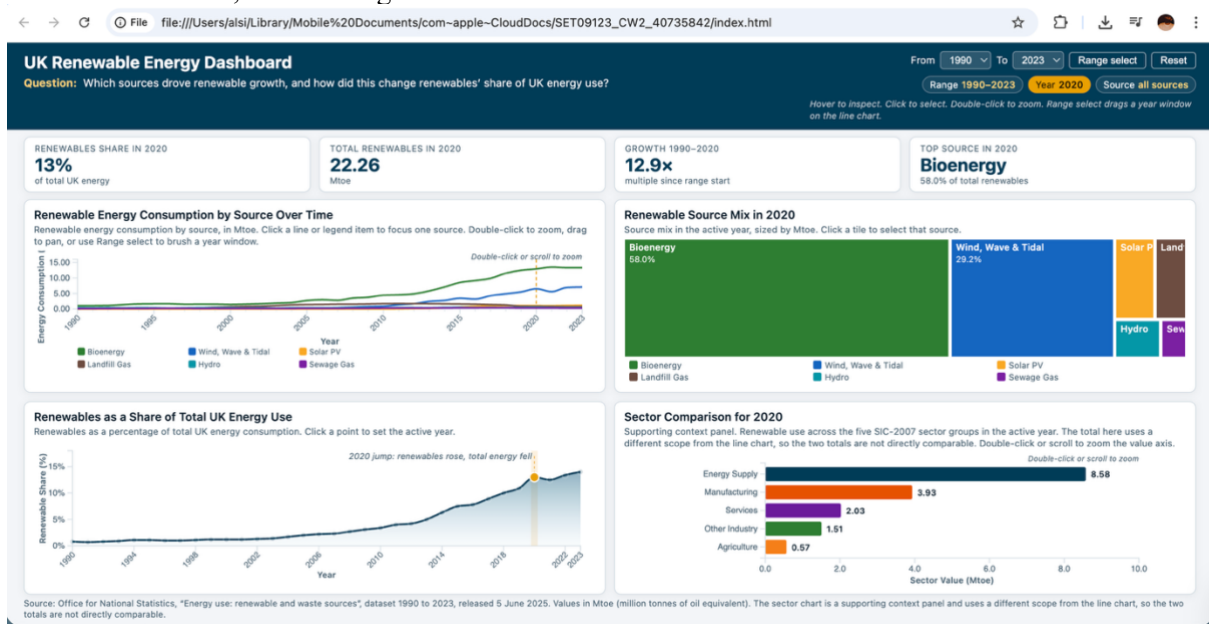


Fig.4- Bioenergy and 2020 are both active.

Double-click on the line chart or the sector chart to zoom in 2 times. Scroll wheel zooms continuously. The Range select button toggles a brush on the line chart, and on release, every chart filters to that range. Reset clears everything in one click. Fig. 5 shows the brush mid-drag.

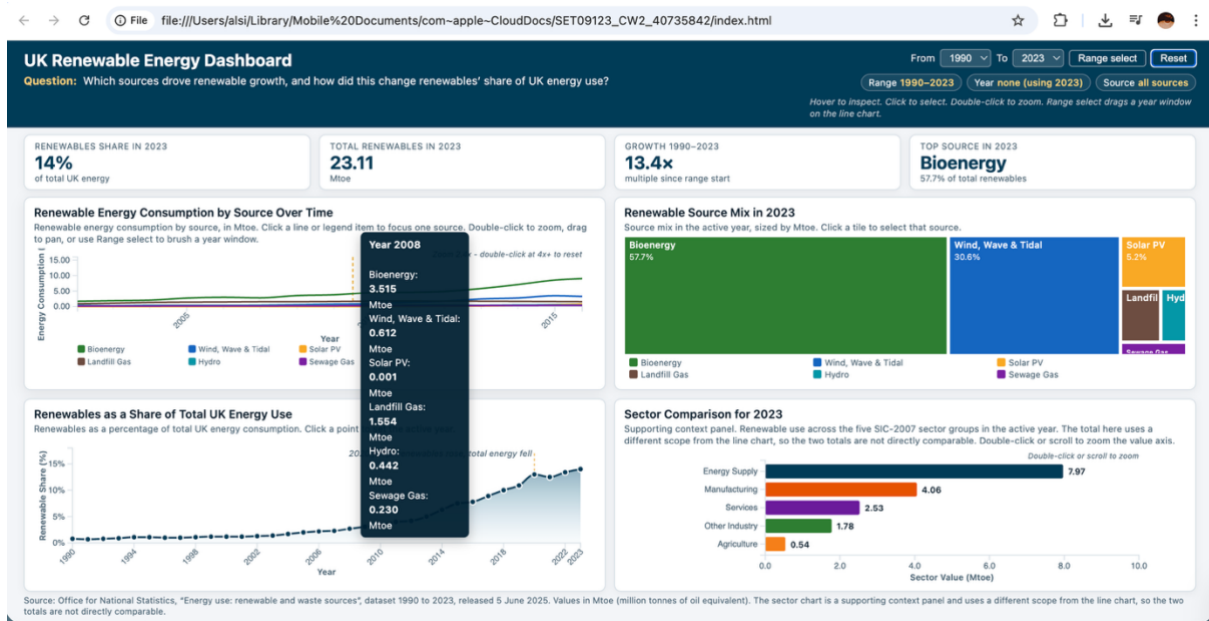


Fig.5- Range select active, brush mid-drag. The button highlights amber.

## 9. The KPIs

A KPI strip showing only totals would be wallpaper. I made the four cards adapt to whatever you have filtered. With no source selected, you see overall renewables share, total renewables, growth multiple, and the top source. With a source selected, the cards switch to that source's value, share of renewables, growth, and the overall renewables share for context.

There is a small subtlety with the growth multiple. For Solar PV, the value at 1990 is zero, because UK Solar PV did not really exist yet. So the growth multiple is mathematically undefined. Instead of showing 'n/a', the code searches forward in the range for the first year where the source crosses 0.01 Mtoe and uses that as the baseline. So Solar PV growth shows around 57 times since 2011, and Wind shows around 370 times since 1993, which are meaningful numbers (the actual values are 56.9 and 372.5, the cards show one decimal). The card label updates to show the actual baseline year.

## 10. Reflection on the interaction design (LO5)

Looking back, a few things in the design ended up working well. Separating hover from click stopped accidental state changes when the cursor moves across four panels, which happens a lot when you are exploring. Independent source and year selection lets you combine filters in two clicks, which was the way I actually wanted to explore the data myself when I was first looking at the spreadsheet. The state is centralised in one object that all charts subscribe to, so I never had to wire chart-to-chart connections, and that turned out to be lucky because it made adding the brush later much simpler than I expected.

Some things did not work as well. The line chart zoom only rescales the time axis. If you zoom in on 2010 to 2020 and read it as the whole story, the trend looks shorter than it really is. The corner shows the zoom level, which helps a bit, but a more careful design would also include a small overview strip below the main plot to anchor the zoomed view. Double click is also not very obvious as a gesture. Touch users have no equivalent because double tap on touch devices is reserved for accessibility, so the dashboard is really desktop only at the moment. Deselecting a source by clicking the same tile again works, but is a bit hidden. The Reset button covers it, but a small, clear button next to each badge would be more obvious.

Given those limits, I leaned toward analytical depth over first-time simplicity. The target audience for this kind of dashboard is people who are already comfortable filtering and combining views, so I think

the trade-off is acceptable. A novice user might be confused for the first 30 seconds. If I had more time I would add a small inline help panel that shows on first load and explains the selection model.

### **11. Limitations and what the dashboard shows**

The line chart folds Bioenergy into one aggregate. Bioenergy is actually a family of sub-sources, including biogas, biomass waste, wood, and liquid biofuels. Splitting them out would be more informative, but the line chart already has six lines, and adding more would push past what people can track by colour. I noted the aggregation in the README and in the hover tooltips. The sector chart uses 5 high-level SIC-2007 categories instead of the full 22 because a 22-row layout would not fit the panel, and many of those rows are near zero anyway.

Three findings come out of using the dashboard. First, Bioenergy and Wind together drove almost all of the renewable growth from 1990 to 2023, with Bioenergy at 57.7% of UK renewables in 2023 and Wind at 30.6%. Second, the renewable share of total UK energy went from under 1% to 14%, which is roughly a 17 fold rise. This was driven by 13.4 times more renewable consumption combined with a 27% fall in total UK energy use over the period. Third, Energy Supply absorbs the bulk of renewable use in every year, and the gap to Manufacturing and Services has actually widened over time, which I did not expect when I started looking at the data.

The dashboard runs from a single HTML page with a small CSS file and five JavaScript files. It works straight from the file system because all data is embedded inside data.js. I tested in Chrome 121, Firefox 122, Edge 121, and Safari 17.

### **References**

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